

FACULTY OF ECONOMIC AND MANAGEMENT SCIENCES/ FAKULTEIT EKONOMIESE EN
BESTUURSWETENSKAPPE

DEPARTMENT OF ECONOMICS/ DEPARTEMENT EKONOMIE

Module Code: 12084-114

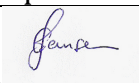
ECONOMICS 114/ EKONOMIE 114

ASSESSMENT A2 / ASSESSERING A2

CLOSED BOOK/ TOE BOEK

PAPER 1 OF 1 / VRAESTEL 1 VAN 1

DATE/ DATUM	23 May 2024/ 23 MEI 2024
DURATION/TYDSDUUR	3 HOURS / 3URE
TOTAL OF MARKS/ TOTAAL PUNTE	100
EXAMINERS/EKSAMINTORE:	DR C LINKS PROF S DU PLESSIS MS O EZEObI MR M MEYER DR K MALINDI MR J PRETORIUS DR D SHEPHARD DR P VORSTER
INTERNAL MODERATOR/ INTERNE MODERATOR	DR E MOSES

Chairperson:	Co-ordinator:
Prof AI Jansen 	DR C LINKS

INSTRUCTIONS

PLEASE NOTE: ONLY NON-PROGRAMMABLE CALCULATORS ARE ALLOWED.

ANSWER **SECTION A's** QUESTIONS ON THE SUPPLIED **ANSWER SHEET**. INDICATE YOUR CHOICE BY **DARKLY COLOURING THE CIRCLE** NEXT TO THE NUMBER OF THE STATEMENT.

ALSO ANSWER **SECTION B's** QUESTIONS ON THE SUPPLIED **ANSWER SHEET**. INDICATE YOUR CHOICE BY **DARKLY COLOURING THE CIRCLE** NEXT TO THE NUMBER OF THE STATEMENT.

ANSWER **SECTION C's** QUESTIONS IN THE ANSWER BOOK PROVIDED. START EVERY QUESTION ON A NEW PAGE. SHOW ALL CALCULATIONS.

KEY:

AP = Average product
AC = Average cost
BL = Budget line
D = Demand
DWL = Deadweight loss
FF = Feasible frontier
GDP = Gross domestic product
IC = Indifference curve
K = Capital
L = Labour
MB = Marginal benefit
MC = Marginal cost
MEB = Marginal external benefit
MP = Marginal product
MPB = Marginal private benefit
MPC = Marginal private cost

MR = Marginal revenue
MRS = Marginal rate of substitution
MRT = Marginal rate of transformation
MSB = Marginal social benefit
MSC = Marginal social cost
P = Price
PF = Production function
Q = Quantity
 Q_d = Quantity demanded
 Q_s = Quantity supplied
S = Supply
TC = Total cost
TP = Total product
TR = Total revenue
w = Wage
Y = Income

SECTION A
TRUE OR FALSE STATEMENTS

[16 marks: 1 mark for a correct answer and - ½ for an incorrect answer]

1. The game shown below is an example of the Prisoner's Dilemma.

		Player 2	
		Vaccinate	Don't vaccinate
Player 1	Vaccinate	5, 5	2, 6
	Don't vaccinate	6, 2	3, 3

- a) True
b) False
2. In the Invisible Hand game, each player's rational, self-interested behaviour creates a benefit for the other player.
a) True
b) False
3. A Pareto improvement occurs when a change from an initial allocation of goods makes at least one person better off.
a) True
b) False
4. Social dilemmas are defined as Nash equilibria that are Pareto efficient.
a) True
b) False
5. Microsoft is owned by many shareholders. The role of these shareholders is to keep a close eye on the performance of the manager.
a) True
b) False
6. Theresa works for a company in Stellenbosch that has recently increased her housing and medical insurance benefits. Under such conditions Theresa will have a high employment rent, ceteris paribus.
a) True
b) False

7. The price elasticity of demand for the product of a price-setting firm at a price of R50, is 1,5. This implies that a 10 percent increase in price will lead to a decrease in total revenue.
 - a) True
 - b) False
8. The competitive market equilibrium price is that price which results in the quantity supplied being equal to the quantity demanded.
 - a) True
 - b) False
9. Firms prefer to enter competitive markets where zero economic profit is being made by existing firms.
 - a) True
 - b) False
10. If $MR = MC$, the quantity produced by a firm is always Pareto efficient.
 - a) True
 - b) False
11. If government wants to raise more revenue through tax, they should levy a tax on a product that has a price inelastic demand.
 - a) True
 - b) False
12. A price-taking firm has a horizontal marginal revenue curve that lies above the demand curve.
 - a) True
 - b) False
13. For a profit-maximising monopolistic firm, $P = MC$.
 - a) True
 - b) False
14. Unregulated price-setting firms can earn positive economic profits in both the short and long run.
 - a) True
 - b) False

15. The toxic material that is used in the production of polyester fibre leads to water pollution. Supposing that the market for polyester fibre is competitive, an efficient quantity of polyester fibre can be attained if a tax is set equal to the marginal external cost of pollution at the efficient equilibrium.

- a) True
- b) False

16. In an unregulated competitive market, the presence of negative externalities will lead to overproduction.

- a) True
- b) False

SECTION B MULTIPLE CHOICE QUESTIONS

[44 marks: 2 marks for a correct answer and – ¼ for an incorrect answer]

17. Consider the following game scenario, where Zendaya and Timothée must each choose a movie to watch at the cinema:

		Zendaya	
		Call me by your Name	Dune
Timothée	Call me by your Name	4, 1	3, 2
	Dune	2, 0	6, 5

Which one of the following options is **CORRECT**?

- a) Timothée has a dominant strategy, because he always gets a higher payoff than Zendaya.
- b) Zendaya has a dominant strategy, which is to choose "Call me by your Name".
- c) Both players have a dominant strategy, which is to choose "Dune".
- d) None of the above.

18. Pam and Debbie must choose between going to the cinema or watching football. The matrix below represents their level of happiness (payoff), depending on the choice of activity. Given this information, which one of the following statements is **TRUE**?

		Pam	
		Football	Cinema
Debbie	Football	3, 5	1, 1
	Cinema	4, 3	6, 2

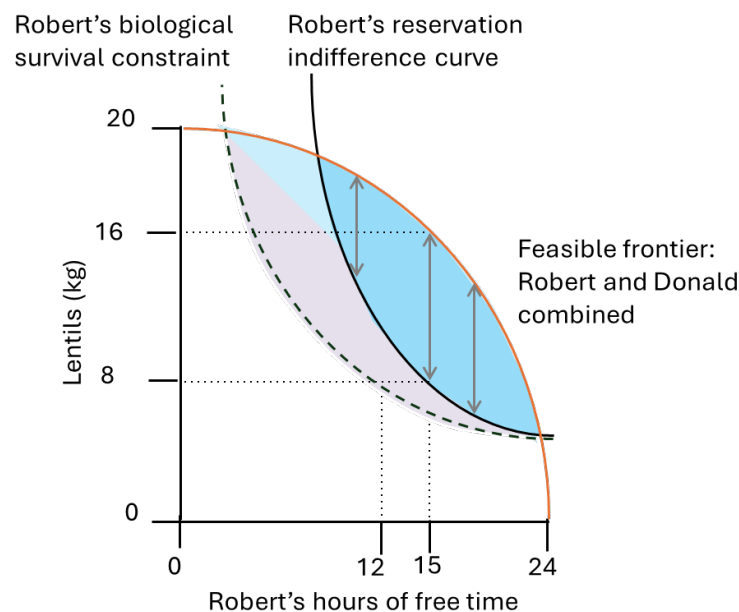
- The dominant strategy for both players is to watch football.
 - The dominant strategy equilibrium is Pareto inefficient.
 - Neither player would want to deviate from the dominant strategy equilibrium.
 - The dominant strategy equilibrium yields each player's best possible outcome.
19. In a soccer game, the two team captains, Verstappen and Le Clerc, are considering whether to rest or not to rest various members of their teams. The game has the following payoff matrix:

		Verstappen	
		Rest	Don't rest
Le Clerc	Rest	0, 0	-1, 1
	Don't rest	1, -1	-2, -2

Which one of the following options is **FALSE**?

- This game has more than one Nash Equilibrium.
 - The best response for each team captain is to do the opposite of what the other captain chooses.
 - Both should avoid resting to gain time on the pitch, despite the risk.
 - None of the captains has a dominant strategy.
20. Which of the following will result in a higher employment rent of a job?
- The existence of another job that pays a higher wage.
 - Employer provision of onsite childcare.
 - Unemployment benefits that are only paid for a portion of the expected duration of unemployment.
 - Higher effort costs of the job.
- Only I
 - II and III
 - III and IV
 - Only IV

21. Five friends must decide how to allocate a chocolate cheesecake among themselves. An allocation in which each friend receives a fifth of the cake is Pareto _____, while an allocation in which each friend receives a sixth of the cake with the remainder of the cake to still be allocated is Pareto _____.
- efficient; inefficient
 - efficient; efficient
 - inefficient; efficient
 - inefficient; inefficient
22. Donald is a landowner and Robert is a lentil farmer. Robert pays Donald a share of his lentil output as rent for the land.



Which one of the following statements is **CORRECT**?

- Under a take-it-or-leave-it rent agreement, Robert will work for 12 hours and keep 2kgs of lentils.
- Under a take-it-or-leave-it rent agreement, Robert will only work 9 hours if he receives more than 10 kgs of lentils.
- Under a take-it-or-leave-it rent agreement, any allocation in the economically feasible set will result in a Pareto improvement over one where Robert and Donald do not reach an agreement.
- Donald's share of the lentils will be maximised at the point where Robert's biological survival constraint curve intersects the feasible frontier.

23. Which of the following factors can increase a worker's employment rent?

- I. The costs of searching for a job
- II. Negative work conditions
- III. An increase of firm-specific assets
- IV. The social stigma associated with unemployment

- a) I and II
- b) III and IV
- c) I, II and III
- d) I, III and IV

24. Which one of the following statements could be a reason for why large firms may be faced with a principal-agent problem?

- a) Workers and managers all have the same information.
- b) Larger firms usually have more market power.
- c) Managers do not necessarily strive to maximise profits.
- d) It is possible to monitor the actions of workers at all times.

25. Ken works in a job that pays a wage of R80 an hour, and his work week is 35 hours long. He exerts an effort level of 0.6 and experiences disutility of work equal to R15 an hour. Assume that it would take him 5 weeks to find a new job should he become unemployed. During that time, he can claim unemployment benefits of R25 per hour for the full duration of his unemployment. Using this information, Ken's total employment rent is _____.

- a) R5 000
- b) R6 200
- c) R7 000
- d) R7 300

Use the following average cost function to answer Question 26 and Question 27:

$$AC = 180\,000/Q + 32 + 2Q$$

26. The total cost of producing 200 units is:

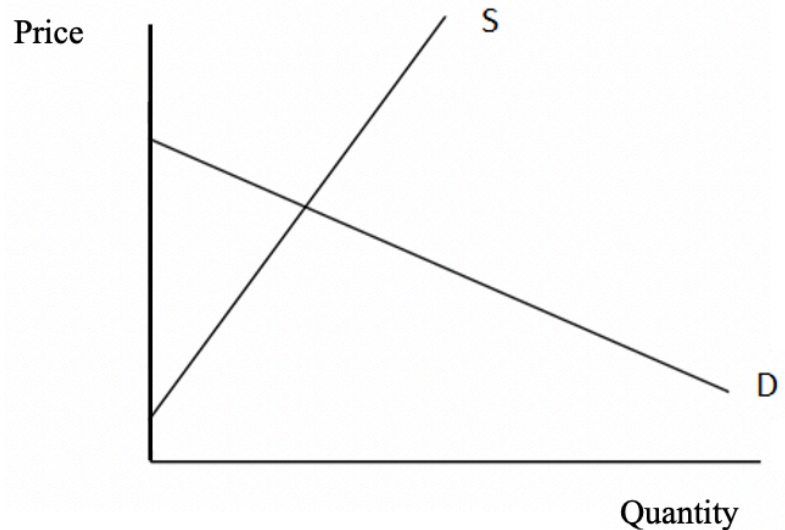
- a) 1 332
- b) 34
- c) 266 400
- d) 832

27. The marginal cost of producing the 200th unit is:

- a) 832
- b) 1 332
- c) 34
- d) 266 400

28. The market for dark chocolate has many sellers and buyers, all of whom are price takers. A global study finds that dark chocolate has significant health benefits, including reducing inflammation and decreasing the risk of heart disease. Which one of the following statements is **CORRECT**?
- a) The market supply curve will shift to the right and the market demand curve will shift to the left. As a result, the equilibrium quantity will increase.
 - b) The market demand curve will shift to the right, the equilibrium price of dark chocolate will increase, and the equilibrium quantity will decrease.
 - c) The market supply curve will shift to the right and the market demand curve will shift to the right. As a result, the equilibrium price of dark chocolate may increase, decrease, or remain unchanged.
 - d) The market demand curve will shift to the right, the equilibrium price of dark chocolate will increase, and the equilibrium quantity will increase.
29. Each firm in a perfectly competitive market
- a) can influence the price of its product.
 - b) produces a product that differs slightly from the products of its competitors.
 - c) is a price taker.
 - d) has a perfectly inelastic demand for its product.
30. Which one of the following statements is **CORRECT**?
- a) The deadweight loss will be larger when a price-setting firm faces elastic demand, ceteris paribus.
 - b) The deadweight loss will be larger when a price-setting firm is able to set a larger price mark-up, ceteris paribus.
 - c) A tax will be effective to dissuade use of a "bad" product if the product has relatively inelastic demand.
 - d) Consumer surplus will always decrease when a binding price ceiling is introduced.

31. The graph below shows the demand and supply curves of a certain good. A per unit tax is imposed on this good.



Which ONE of the following statements regarding tax incidence is **CORRECT**?

- a) The tax incidence on buyers will exceed the incidence on sellers because demand is relatively less price elastic than supply.
- b) The tax incidence on buyers will exceed the incidence on sellers because demand is relatively more price elastic than supply.
- c) The tax incidence on sellers will exceed the incidence on buyers because demand is relatively more price elastic than supply.
- d) The tax incidence on sellers will exceed the incidence on buyers because demand is relatively less price elastic than supply.

Use the following information to answer Question 32 and Question 33:

A firm with market power has the following demand and marginal cost functions:

$$P = 100 - 2Q$$

$$MC = Q$$

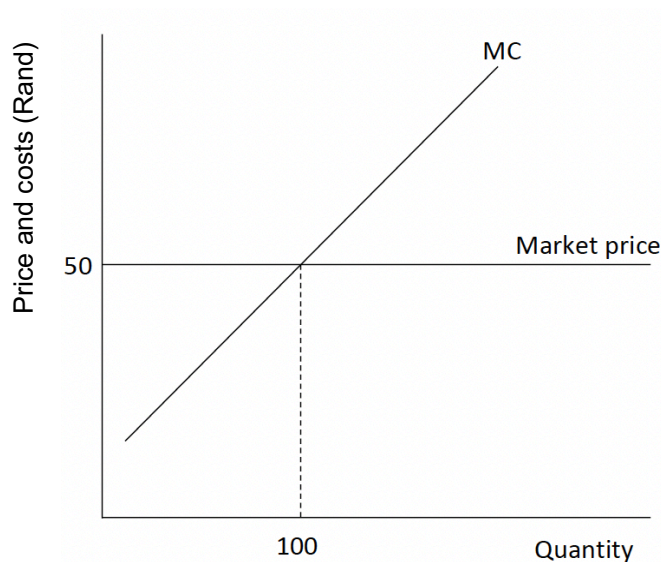
32. What price should the firm set to maximise profits?

- a) R36,67
- b) R60
- c) R52
- d) R40

33. What is the marginal revenue of the fifth unit sold?

- a) R10
- b) R20
- c) R60
- d) R80

34. The figure below shows the marginal cost of production and the market price charged by a firm in a perfectly competitive market in long-run equilibrium.



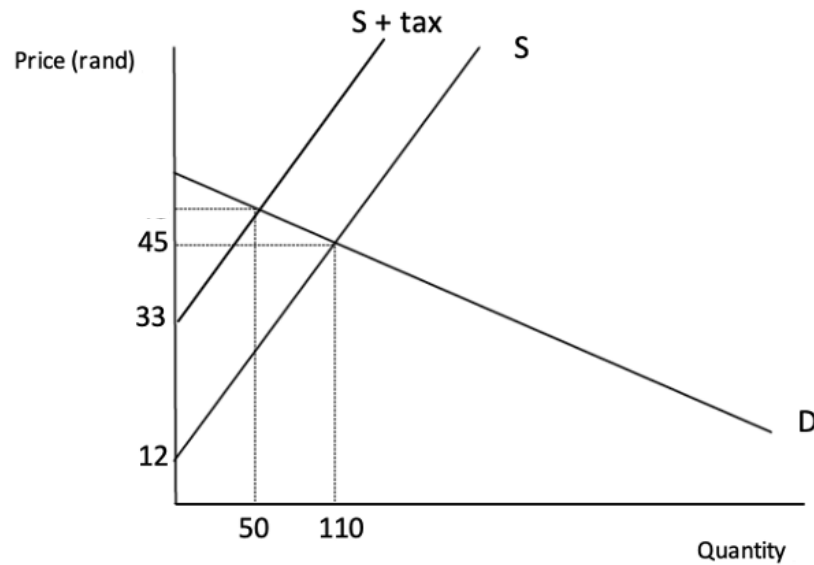
The firm's average cost of producing 100 units is:

- a) R50
- b) R40
- c) R25
- d) R5

35. The market for cucumbers is perfectly competitive. Higher consumer demand for cucumbers results in short-run economic profits for the sellers of cucumbers. Over the long run, the price of cucumbers will _____ and the profit-maximising quantity of cucumbers produced and sold by an individual seller of cucumbers will _____.

- a) fall; increase
- b) rise; decrease
- c) rise; increase
- d) fall; decrease

36. The graph below shows the demand and supply curves of a certain good. A per unit tax is imposed on this good, which results in total sales revenue amounting to R2 400.



The price paid by consumers following the implementation of the tax is _____.

- a) R48
 - b) R31
 - c) R27
 - d) R29
37. Live performances of the musical Wicked where no cell phone recordings are allowed are examples of:
- a) rival and excludable goods.
 - b) rival and non-excludable goods.
 - c) non-rival and excludable goods.
 - d) non-rival and non-excludable goods.
38. The education you receive at university has a positive effect on other members of society. Which one of the following statements is **CORRECT**?
- a) If nothing is done in an unregulated market for higher education, higher education will be undersupplied if universities are private enterprises.
 - b) A socially optimal provision of higher education could be achieved through taxing suppliers of higher education.
 - c) To ensure that the socially optimal quantity of higher education is provided, MSB must always exceed MPB.
 - d) Providing subsidies to the suppliers of higher education will decrease the quantity of higher education supplied.

SECTION C
DESCRIPTIVE QUESTIONS
[40 marks]

Question 39 [4 marks]

Zolani is an indie rock musician who has a back-up band called Freshlyground. One night in the recording studio, Zolani and her bandmates get into an argument and need to decide whether they are going to split up or stay together as a band. Their interaction is represented by the payoff matrix below:

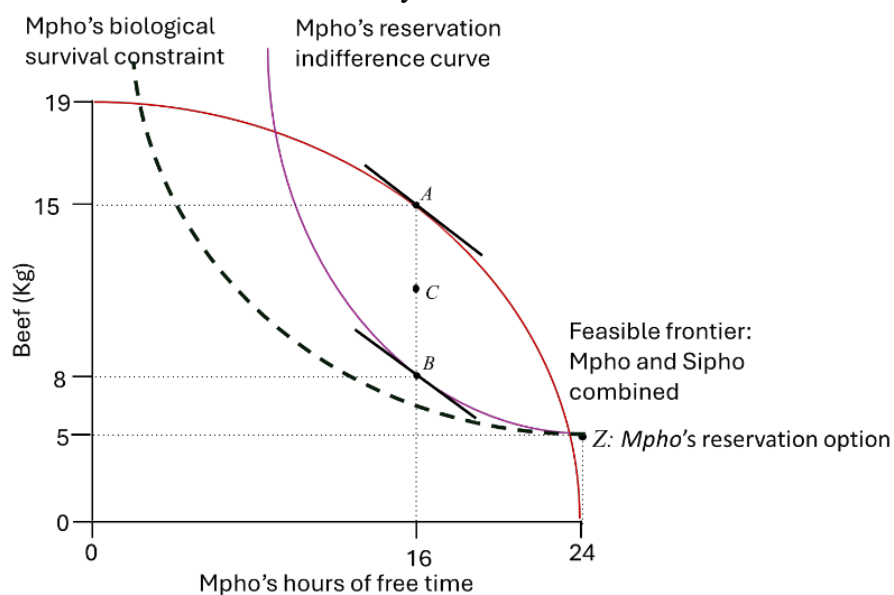
		Freshlyground	
		Stay Together	Split Up
Zolani	Stay Together	5, 5	2, 7
	Split Up	7, 2	3, 3

39.1 List all the Nash Equilibria of this game. [2]

39.2 What is the socially optimal outcome of this game? Motivate your answer briefly. [2]

Question 40 [3 marks]

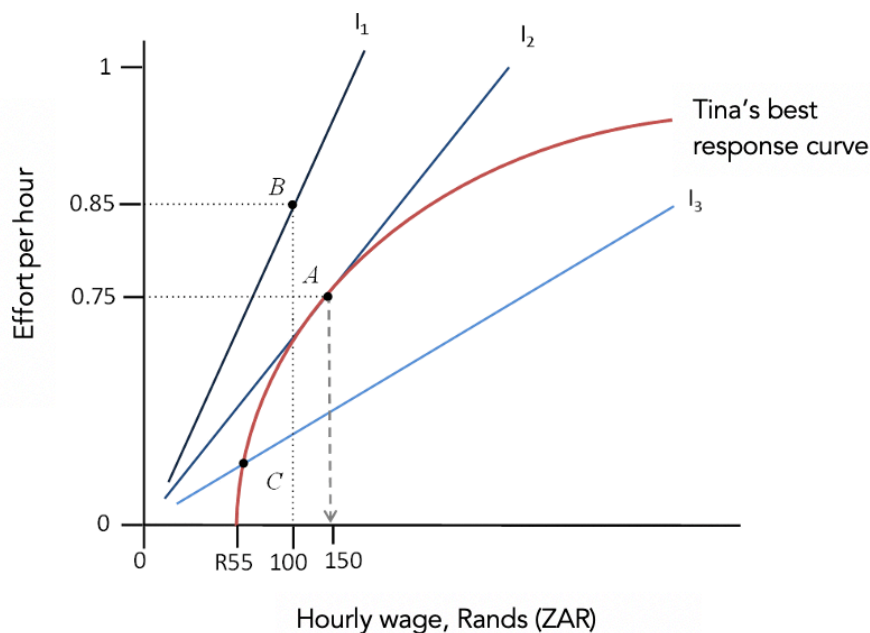
Mpho is a cattle farmer who grazes his cattle on Sipho's farm. He must pay Sipho a portion of his beef for the right to do so. The feasible frontier of Mpho and Sipho as well as Mpho's reservation indifference curve are portrayed in the figure below. Currently, the allocation of beef is according to point **B**, where Mpho pays Sipho 7kg of the total 15kg of beef produced, and Mpho decides to work 8 hours a day.



- 40.1 Does allocation **C** Pareto dominate allocation **B**? Explain. [2]
- 40.2 Under what conditions would the allocation of beef between the two parties move to point **C**? [1]

QUESTION 41 [5 marks]

The figure below indicates the best response curve of Tina, a florist, and several isocost of effort curves of her employer, Blossom Bazaar Floral Trading.



- 41.1 Which one of the isocost curves I_1 , I_2 and I_3 , represents the firm's highest cost of effort? [1]
- 41.2 How many efficiency units per rand are represented by isocost curve I_1 ? [1]
- 41.3 Would it be a feasible option for the firm to pay R55 per hour? Justify your answer. [1]
- 41.4 Which of the points A, B and C would be the most preferred for Blossom Bazaar Floral Trading? Briefly explain. [1]
- 41.5 What hourly wage should the firm pay Tina? [1]

QUESTION 42 [3 marks]

The market for short-term lease bachelor flats, which is competitive, has an inverse demand function given by:

$$P = 10\,000 - 0,25Q_d$$

and an inverse supply function given by:

$$P = 0,15Q_s + 400$$

Government decides to introduce a price ceiling of R7 000 in this market. Is this price ceiling binding? Justify your answer fully. [3]

QUESTION 43 [6 marks]

Susan sells hamburgers at a Sunday market. The demand for her hamburgers is given by:

$$Q_d = 50 - 0,4P$$

She currently charges R40 per hamburger.

43.1 Calculate the price elasticity of demand for Susan's hamburgers. Show all calculations. [4]

43.2 Would you advise Susan to lower the price of her hamburgers? Motivate your answer. [2]

QUESTION 44 [13 marks]

The equations of the market demand and supply curves for laptop sleeves are given by:

$$P = 300 - 0,02Q_d$$

$$P = 60 + 0,01Q_s$$

44.1. Calculate the market equilibrium price and quantity of laptop sleeves. [3]

Suppose the government introduces a R30 per unit tax on the production of laptop sleeves to generate revenue.

44.2 What is the price and quantity of laptop sleeves sold following the imposition of the tax? Show all calculations. [4]

44.3 Draw a fully labelled graph to illustrate the:

- consumer surplus;
- producer surplus; and
- deadweight loss

in the market for laptop sleeves following the imposition of this tax. [5]

44.4 What is the total tax revenue collected by the government? [1]

QUESTION 45 [6 marks]

The market for orange juice is unregulated and perfectly competitive. Suppose the marginal private cost of producing a crate of orange juice is:

$$P = 100 + Q$$

and that the demand curve for orange juice is given by:

$$P = 600 - \frac{1}{3}Q_d$$

where prices and costs are in Rands. Suppose that the production of orange juice causes pollution that leads to a marginal external cost of $2Q$.

45.1 What is the Pareto efficient quantity (crates) of orange juice? Show your calculations. [2]

45.2 What is the tax revenue from the imposition of an optimal Pigouvian tax? Show your calculations. [2]

45.3 How can private bargaining solve the problem of an external cost that is created during production? [2]